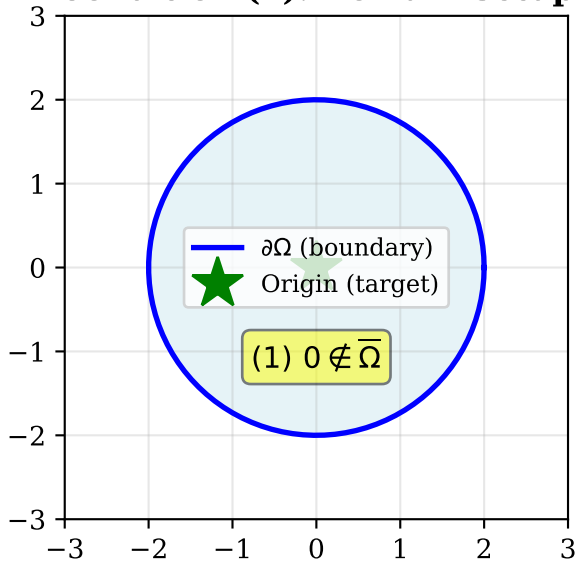


Skrypnik's Degree Theory: $\Phi = I - A$, $A: \bar{\Omega} \rightarrow X^*$

Condition (1): Domain Setup



Condition (2): Operator A properties

- Completely continuous on $\bar{\Omega}$
- Pseudomonotone or demicontinuous
- $\|Au\|_{X^*} \geq \delta_0 > 0$ for $u \in \partial\Omega$

These ensure $\Phi = I - A$ satisfies the $\alpha(\partial\Omega)$ condition

Skrypnik's Theorem: If conditions (1) and (2) hold, then $\deg(A, \bar{\Omega}, 0)$ is well-defined and possesses properties similar to finite-dimensional degree. Key properties:

- Existence: $\deg(A, \Omega, 0) \neq 0 \Rightarrow \exists u \in \Omega$ with $Au = u$
- Homotopy Invariance: Continuous deformation preserves degree (if certain conditions hold)
- Fixed Point: $\deg(I - A, \Omega, 0) \neq 0 \Rightarrow \exists x \in \Omega: Ax = x$